Prototype-Guided Spatial–Spectral Interaction Network for Hyperspectral Anomaly Detection

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***Abstract*— In recent years, deep learning (DL) has emerged as one of the most widely utilized techniques in hyperspectral anomaly detection (HAD) with an impressive detection accuracy. However, the investigation into the diverse backgrounds (BKs) representation and the spatialspectral interaction remains underexplored. To tackle this, we propose a novel framework namely the prototype-guided and spatialspectral interaction network (PSSIN) for HAD in this article. Specifically, an adaptive anomaly mask module (AAMM) is utilized to mitigate the interference of the BK reconstruction caused by the blending of potential anomalies. Subsequently, we design a BK-guided prototype autoencoder (BP-AE) to represent the BKs with various land cover types, incorporating two critical components: the BK prototype module (BPM) and the spatialspectral interaction block (SSIB). To characterize different typical BK features by a global perspective, BPM utilizes a prototype learning strategy with a self-attention mechanism, and a multivariate ensemble loss is employed for BPM to optimize the transformation of BK features and the updating of a prototype codebook. To enhance the spatialspectral utilization of window-based approach, SSIB first introduces a spatialspectral interaction paradigm for HAD. The window-based self-attention branch is to mine spatial feature characteristics, while the depth-wise convolution branch is to extract spectral features. These two branches in a parallel configuration interact with each other's features and then perform feature fusion. SSIB architecture not only broadens the receptive fields by concurrently modeling the intrawindow and cross-window relationships but also facilitates bi-directional interactions between the spatial and spectral branches. Furthermore, the comprehensive experiments conducted on six authentic datasets have fully validated its superior performance.**

***Index Terms*— Hyperspectral anomaly detection (HAD), multivariate background (BK) representation, prototype learning, spatial–spectral interaction.**

I. Introduction

HYPERSPECTRAL image (HSI) remote sensing [1], as an emerging technology and essential instrument, has been successfully applied in a wide range of earth observation applications, such as environmental monitoring, climate change, agricultural management, and urban planning. With the rich spectral information along with the spatial image data, HSI has enabled more accurate remote sensing earth observation in numerous tasks [2], [3], including HSI classification-based land mapping [4], [5], super-resolution [6], [7], and target detection [8].

As an unsupervised computer vision task, hyperspectral anomaly detection (HAD) [9], [10], [11] aims to capture anomaly targets that differ significantly from the surrounding spatial features [12]. These anomalies can be found on the land, ocean, and even space, such as landslide, collapsed bridge, volcano, and star bursting, resulting in critical consequence to people's lives and the economic conditions. Automatically identifying such events is thus vital for disaster assessment and emergency service [13]. To this end, a multitude of diverse HAD algorithms have been developed in Table I, which can be broadly categorized into two types: traditional algorithms and deep learning (DL) algorithms, as detailed below.

A. Traditional HAD Algorithms

In the early days, the Reed–Xiaoli (RX) [14], a statistical-based method, was a pioneering work in the HAD field, based on the assumption that the HSI background (BK) conforms to a Gaussian multivariate distribution. RX is based on the assumption that the HSI BK conforms to a Gaussian multivariate distribution. Due to the presence of anomalies and noise mixed into the BK samples in RX, there is a problem with high false alarms in the detection maps. To deal with it, a subspace RX [15] is introduced, wherein multiple BK dimensions characterized by high variance are eliminated to mitigate the impact of anomalous pollution on BK construction. Similarly, weighted-RX [16] assigns low weights to potential anomalous pixels and high weights to potential BK pixels. In addition, some other versions based on RX are proposed for HAD to improve detection accuracy, such as iterative RX [17].

Representation-based algorithms include three types: sparse representation (SR), collaborative representation (CR), and low-rank representation (LRR). In the SR models, the fundamental premise posits that normal pixels can be effectively represented by the constructed overcomplete dictionary atoms, while anomalous pixels are not amenable to such representation. To extract physical significance, a constrained SR

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TABLE I
TAXONOMY OF HAD METHODS

Category	Subcategory	Methods
Traditional based	Statistics[14-17]	RX (1990)[14], SRX(2004)[15], LAIRX (2010)[17], WRX(2014)[16]
	SR[18-20]	SSEAD(2017)[19], SR-LMM(2018)[18], MDSR(2018)[20]
	CR[21-23]	CRD(2015)[21], NJCR(2022)[22],SRCR(2023)[23]
	LRR[24-26]	DCC-LRSR-RDC(2023)[25], DGRAD-LRR(2024)[24], ELRSF-SP(2024)[26]
	Other[27-32]	Filter based: AED(2017)[27] Cluster based: QPAD(2022)[28] IF based: KIFD (2022)[29], MSGDF (2022)[30], DFTSIF(2023)[31] Tensor bsed: GNLTR (2023)[32]
Deep Learning based	GAN[35-38]	GAN-SSL(2020)[35], WSLRR(2021)[36], FTSGAN(2023)[37], CL-CaGAN(2024)[38]
	AE[39-51]	RGAE(2021)[39] Memory based: OTCMA(2022)[40], MAAE (2024)[41] AE with LRR: DSLF(2024)[42], DLRSPs-DAEs(2024)[43] Pix Relationship based: GAED(2022)[44], PAAE(2024)[45], LCNN-RLBE(2024)[46] Blind-Spot based: BockNet(2023)[47], NAFNet(2024)[48], DirectNet (2024)[49] Model Interpretation: LRSR-I2Net (2024)[50], LRR-Net(2023)[51]
	Other[52-53]	AGFL(2024)[52], TDD(2024)[53]

approach [18] grounded in a linear mixture strategy has been introduced for HAD. Zhao et al. [19] designed an estimated sparsity score, which assesses the frequency and intensity of the learned dictionary. The CR posits that the collaborative interactions among all atoms are more crucial than the competitive dynamics of individual atoms [20]. In the CR, it is anticipated that each standard pixel can be represented by BK dictionary elements, while anomalous pixels remain unattainable. The initial CR [21] employs a linear combination utilizing a sliding dual window. A novel constrained regression model featuring a nonnegative constraint and joint learning approach is introduced for HAD [22]. Lin et al. [23] developed a novel ensemble approach based on SR and CR to improve the representation ability. The SR and CR methodologies overlook low-rank characteristics present in the BK elements. To address this issue, the LRR model segregates an HSI into the low-rank BK and the sparse anomaly. Thereby, the LRR is employed for HAD. In terms of mining the spatial and spectral structure, a DGRAD-LRR [24] integrates a dual graph regulation into LRR with the adaptive dictionary strategy. Furthermore, alongside these LRRs, there exist several variants specifically engineered to improve the LRR model [25], [26].

In addition, there are also other conventional techniques for HAD, such as the filter-based approach [27], the cluster-based technique [28], the isolation forest (IF) methods [29], [30], [31], and the tensor-based strategy [32].

B. DL HAD Algorithms

DL has emerged as a leading technology in recent years, extensively utilized in computer vision [33], [34]. Consequently, it is also applied in HAD. Despite HAD lacking prior knowledge, it leverages the fact that the BK constitutes a substantial majority of the image, while anomalies occupy a minimal area. Thus, DL models exploit the disparities between BK and anomaly representations to effectively identify anomalies. From the perspective of model structure, current DL methods in the HAD can be roughly divided into two types: the generative adversarial network (GAN) and the autoencoder (AE).

In the GAN-based HAD, the most notable contribution is originated from an innovative GAN-driven semi-supervised spectral learning (SSL) approach [35], aiming at thoroughly leveraging the complementary BK information inherent in both the spectral and spatial features. Considering the impact of the effectiveness of annotations on model performance, a weakly supervised GAN Xie et al. [36] is proposed with a weakly supervised BK estimation model and a dictionary-based LRR integrated for HAD. To tackle the noise and limited spatial resolution, a frequency-to-spectrum mapping GAN in a semi-supervised manner Wang et al. [37] is proposed to deal with it. Wang et al. [38] introduce a novel capsule differential adversarial continual learning framework to improve cross-scenario learning capabilities, thereby advancing the practical application of DL-based structures in the HAD.

The AE is recognized as a prevalent unsupervised model that demonstrates commendable efficacy within the HAD domain. Among them, a novel robust graph AE [39] is introduced, incorporating the $L_{2,1}$ norm and superpixel graph regularization. This approach can preserve the geometric structure. In [40], a cluster-memory augmented deep AE with optimal transportation is proposed to learn the consistency and differentiation among features and weaken the anomaly reconstruction ability. Likewise, an adaptive reconstruction and sample attribution mining strategy is designed to improve memory-augmented AE [41]. Some scholars combine the low-rank SR with AE to build an end-to-end manner, such as DSLF [42] and DLRSPs-DAEs [43]. This way can provide the good prior knowledge to enhance the BK reconstruction. The relationship between pixels in the local region is of great significance for HAD. Based on this, several frameworks [44], [45], [46] are put forward. Among them, GAED [44] is a representative work. It adopts the curvature filter technology to measure the similarity between the current pixel and the surrounding eight pixels, and the results are utilized for model optimization. For cases where abnormal pixels are well reconstructed, some blind-spot networks [47], [48], [49] are present to address this issue. The literature [47] constructs a self-supervised blind-block network with a guard window. To fully use of spatial–spectral features, a novel BK reconstruction network is created for HAD called the dual-window-inspired reconstruction Network [49] (DirectNet). This network effectively incorporates a sliding dual-window model within a blind-block structure. Regarding the DL unexplainability [50], LRR-Net creates a link between low-rank model-based approaches and data-driven DL methodologies to achieve model interpretation [51].

In addition to GAN and AE-based models, there are also other DL-based approaches proposed for HAD. For example, Tu et al. [52] introduce an innovative HAD approach that

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leverages the graph frequency analysis in the DL model design, successfully integrating both the structural information of graphs and their spectral characteristics. In light of the intricacies associated with the existing two-stage detection strategies, a pioneering one-step detection paradigm with DL transfer ability is proposed [53] for identifying the spectral deviation relationship based on the defined anomalies.

C. Motivations and Contributions

The existing DL approaches have demonstrated satisfying efficacy in the HAD domain. However, there are two major challenges that warrant additional exploration as detailed below.

First, most DL models treat HAD primarily as a binary classification, with an underlying assumption to consider all BK pixels as one class and the anomalies as another. As the BK in an HSI is complex and consists of various surface features, the assumption may severely affect the detection accuracy. Recent studies [54], [55] suggest that it is essential to consider multiple types of normal data for anomaly detection.

Second, DL models in HAD have increasingly focused on leveraging spatial and spectral information in two different ways: 1) employing separate branches for spatial and spectral learning and detection, followed by decision-level fusion [56]; and 2) fusing the extracted spatial and spectral features before conducting the detection [57]. Although both of these may yield good results, they exhibit limitations regarding the intrinsic nature of HSI data. To be specific, the fusion way at either the feature or decision level is deficient in the complementary and interactive functions between the spatial and spectral features, where their interactions are often neglected [58]. Additionally, an effective integrated spatial–spectral strategy is crucial for model construction [59].

To tackle these two challenges, a prototype-guided and spatial–spectral interaction network (PSSIN) is proposed for HAD in this article. As shown in Fig. 1, we conceive two parts in the proposed framework, namely adaptive anomaly mask module (AAMM) and the BK-guided prototype AE (BP-AE). Specifically, the AAMM is designed to remove potential anomalies and reduce their interference in model training. The mask map and an HSI are fed into the BP-AE in order to construct two modules from both global and local perspectives: the BK prototype module (BPM) and the spatial–spectral interaction block (SSIB). The major contributions of the PSSIN are highlighted below.

1) A novel prototype-guided and spatial–spectral interaction network is proposed for HAD. As is shown in Fig. 1, the BPM is crafted to represent the BKs encompassing multiple land cover classes. By introducing a prototype learning strategy grounded in a self-attention mechanism, it can not only explore the relationships among prototype elements but also reveal the relationship between prototype codebooks and HSI features.
2) A novel spatial–spectral interaction paradigm, SSIB, is first utilized for HAD in our consciousness, as a plug-and-play module to enhance the utilization of spatial–spectral features. By combining local-window self-attention with depth-wise convolution in a parallel configuration, SSIB can not only broaden the receptive fields by concurrently modeling the intrawindow and cross-window relationships but also foster bi-directional interactions in between.
3) Experiments on six HSI datasets have proved the superiority of the proposed PSSIN in eleven state-of-the-art (SOTA) approaches. The ablation study has demonstrated that the PSSIN design is reasonable and effective.

II. Proposed Method

A. Overview

The proposed framework is shown in Fig. 1(a), which has two major parts: AAMM and BP-AE. The BPM is embedded in the middle of the BP-AE to characterize different types of BKs, while the SSIB is added into the encoding layers and decoding layers for mining local information, as detailed below. As illustrated in Fig. 1(b), the BPM is integrated into the BP-AE and leverages a prototype codebook to represent the BK pixels using the global characteristics. The BK in an HSI encompasses various land covers. In the codebook, each prototype item can quantify a BK cover attribute. For the construction of each prototype item, local prior knowledge with mask information is utilized to generate local BK centroids, with each centroid representing a typical feature of a local BK cover. Consequently, each centroid feature is employed to initialize every prototype item. To optimize these prototype items and transform HSI features, a self-attention mechanism with the separation loss is implemented to uncover the similarities among prototype items, and a multivariate ensemble loss function is devised to constrain the HSI features. A feature compactness loss is introduced to explore the intraclass similarity between each HSI feature and the prototype items, while a feature diversity loss is applied to enhance their interclass differentiation. Through the BPM, multiclass BK representation is achieved, thereby improving model sensitivity to anomalies. As the spatial–spectral interaction is beneficial to HSI classification, a local spatial–spectral interaction block is first employed for HAD, as shown in Fig. 1(c). This block consists of two branches designed to extract local spatial features and spectral features independently, facilitating interaction between the spatial and spectral features before fusing them. Meanwhile, different receptive fields are established to dig out local information. In contrast to previous methods of the spatial–spectral utilization, the SSIB design more comprehensively considers the relationship between spatial and spectral information.

B. Adaptive Anomaly Mask Module

Most DL methods in the HAD optimize the BK representation to enlarge the reconstruction errors of anomalies. However, abnormal pixel participation in model training can weaken the differentiation of BKs and anomalies during the test phase. Inspired by [12] and [47], we construct the AAMM in this article. This strategy is based on the anomaly attribute which is that anomalies are rare and different in an HSI. To this end, the density-based spatial clustering of applications with

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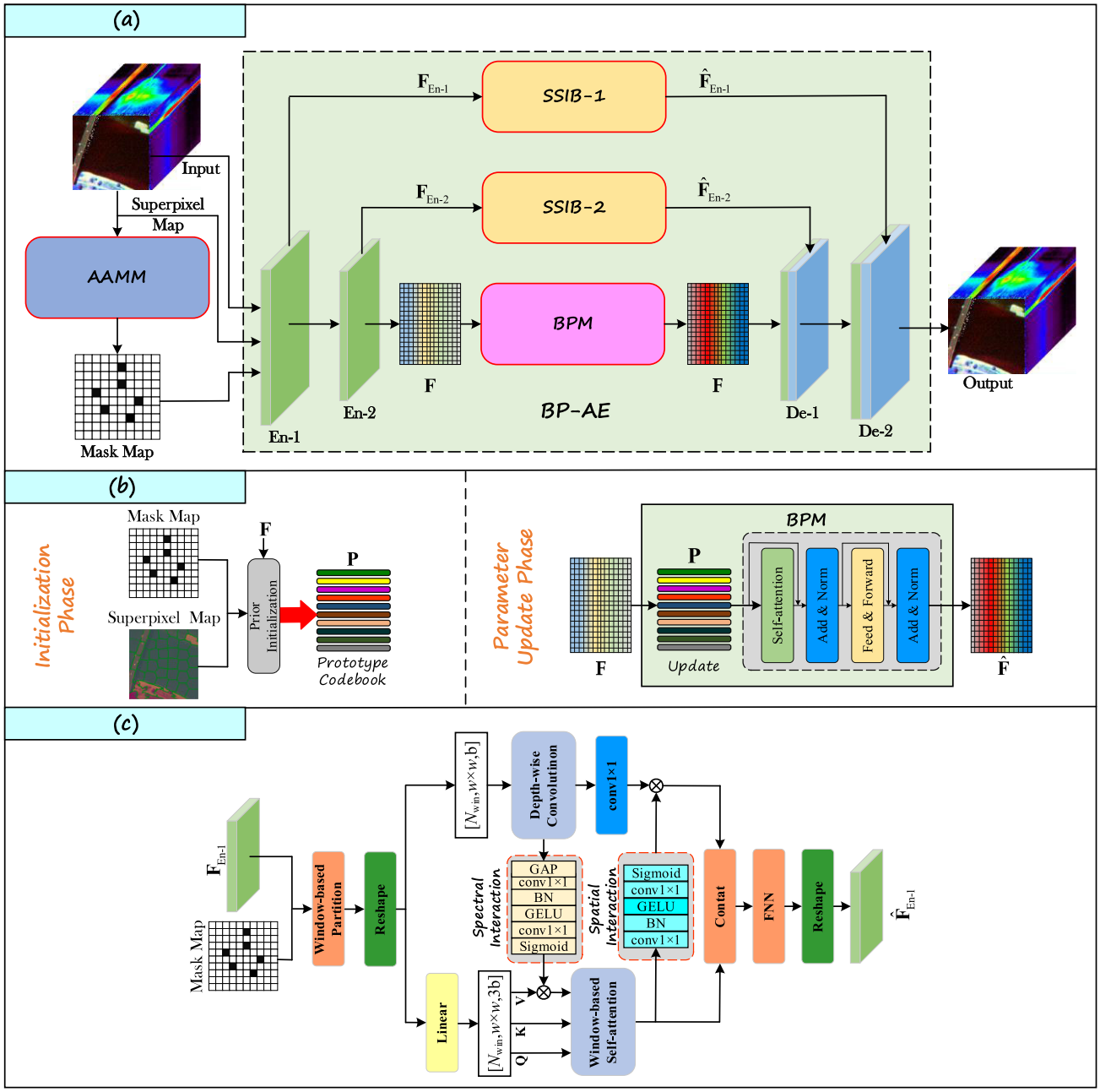


Fig. 1. Framework of the proposed PSSIN, which has two main parts: AAMM and BP-AE. The BPM and SSIB are integrated into the BP-AE. (a) PSSIN Framework. (b) BPM. (c) SSIB.

noise (DBSCAN) is utilized to distinguish diverse types of land features, and it can adaptability adjust the cluster count based on the HSI data. By observing the clustering results of DBSCAN, it is found that classes with relatively fewer pixels are abnormal. Classes with many pixels may be from different spatial regions. If a subset of pixels is aggregated into a big homogeneous area, these pixels can be classified as BK. Conversely, if a collection of pixels is concentrated within a small area, they are likely to represent anomalies. To capture this type of anomaly, a simple yet effective way using the pixel-based eight-connected neighborhoods is adopted, as it can count the number of pixels in different regions within the same class. To this end, the number of pixels from different regions in various clusters is counted. By following [60] to set a reasonable pixel number threshold, we can identify potential anomalies and consider them as mask pixels.

C. BK-Guided Prototype AE

Inspired by [54], it is necessary to consider multiple BKs for unsupervised anomaly detection. Here we propose a prototype learning strategy, which delineates the characteristics of diverse BK elements, thereby encouraging anomalies to be deviated from BKs. In addition, the local information in HAD is important and easily overlooked for DL methods. To enhance the HSI representation, SSIB explores the local similarities within spatial–spectral information by utilizing different receptive fields.

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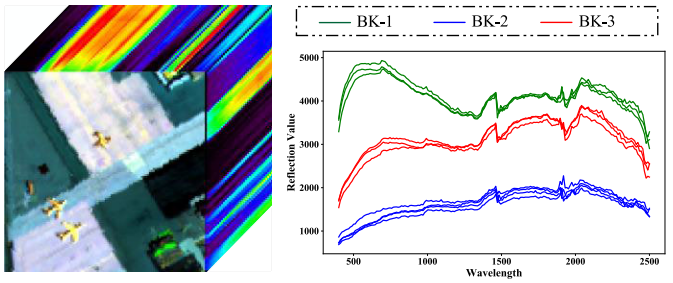


Fig. 2. Spectral characteristics of BK pixels for three different types of land cover, and the same ground object BK pixels have similar spectral curves. Therefore, similar spectral features can be regarded as a type of ground BK.

The BP-AE consists of four parts, i.e., the encoder (En), BPM, SSIB, and decoder (De). An HSI is denoted by $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$; H, W, and B are the length, width, and spectral band, respectively. The BP-AE is defined by

$$\begin{aligned} \mathbf{F}_{\mathrm{En}-1},\ \mathbf{F}_{\mathrm{En}-2},\ \mathbf{F} &= \mathrm{En}(\mathbf{X},\ \mathbf{M},\ \theta_{\mathrm{En}}) \\ \hat{\mathbf{F}} &= \mathrm{BPM}(\mathbf{F}, \mathbf{P},\ \theta_{\mathrm{BPM}}) \\ \hat{\mathbf{F}}_{\mathrm{En}-1} &= \mathrm{Net}_{\mathrm{SSIB}-1}(\mathbf{F}_{\mathrm{En}-1}, \theta_{\mathrm{SSIB}-1}) \\ \hat{\mathbf{F}}_{\mathrm{En}-2} &= \mathrm{Net}_{\mathrm{SSIB}-2}(\mathbf{F}_{\mathrm{En}-2}, \theta_{\mathrm{SSIB}-2}) \\ \hat{\mathbf{X}} &= \mathrm{De}(\hat{\mathbf{F}},\ \hat{\mathbf{F}}_{\mathrm{En}-1}, \hat{\mathbf{F}}_{\mathrm{En}-2}, \theta_{\mathrm{De}}) \end{aligned} \tag{1}$$

where $\mathbf{M}$ denotes a mask map from the AAMM; $\mathbf{F}_{\mathrm{En}-1}$, $\mathbf{F}_{\mathrm{En}-2}$, and $\mathbf{F}$ are features of the encoder layers; among them, $\mathbf{F} \in \mathbb{R}^{N\times b}$ $(N = H \times W)$ is a 2-dimension feature which is transformed from 3-D, where N is the pixel number of an HSI; $\mathrm{En}(\cdot)$ is the encoder function and θ_{En} is its parameter; $\hat{\mathbf{F}}_{\mathrm{En}-1}$ and $\hat{\mathbf{F}}_{\mathrm{En}-2}$ are obtained features by the SSIB, $\mathrm{Net}_{\mathrm{SSIB}-1}(\cdot)$ and $\mathrm{Net}_{\mathrm{SSIB}-2}(\cdot)$ are the SSIB function and they have different window sizes, $\theta_{\mathrm{SSIB}-1}$ and $\theta_{\mathrm{SSIB}-2}$ represent their corresponding parameters; $\mathrm{BPM}(\cdot)$ denotes the BPM function and θ_{BPM} is its parameter; $\mathbf{P}$ is a prototype codebook with a size of $m \times b$ for discrete representation; $\mathrm{De}(\cdot)$ and θ_{De} denote the function and parameter in the decoder. Notably, unlike existing AE-based HAD models, the BP-AE designs the BPM and spatial–spectral interaction module to make full use of the HSI information, which can enhance the multiclass BK representation and weaken the anomaly representation, resulting in high reconstruction errors to highlight anomalies.

Background Prototype Module: In a real HSI of Fig. 2, land features of the BKs are in multiclasses. A prototype learning strategy is utilized for HAD to represent diverse BK pixels. As shown in Fig. 1(a) and (b), the BPM is integrated into the AE, and the optimization of $\mathbf{P}$ and $\hat{\mathbf{F}}$ is described in Fig. 3.

In the stage of transforming $\mathbf{F}$, the input feature is $\mathbf{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_K\} \in \mathbb{R}^{K\times b}$ and $\mathbf{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_M\} \in \mathbb{R}^{M\times b}$ in Fig. 3(a). The similarity matrix between $\mathbf{F}$ and $\mathbf{P}$ is calculated by

$$\mathbf{S}_{k,m} = d(\mathbf{f}_k, \mathbf{p}_m) = \frac{\mathbf{f}_k \mathbf{p}_m^{\mathrm{T}}}{\|\mathbf{f}_k\| \|\mathbf{p}_m\|} \tag{2}$$

where $\mathbf{f}_k \in \mathbf{F}$, $\mathbf{p}_m \in \mathbf{P}$; $\mathbf{S}_{k,m} \in \mathbf{S}$, its size is $K \times M$; $d(\cdot,\cdot)$ is cosine similarity function. Each element $S_{k,m}$ is normalized to $\hat{\mathbf{S}}_{k,m}$, and $\hat{\mathbf{S}}_{k,m}$ is utilized to convert $\mathbf{F}$ to $\hat{\mathbf{F}}$ as follows:

$$\hat{\mathbf{S}}_{k,m} = \frac{\mathbf{S}_{k,m}}{\mathbf{S}_{I_{\max}}}, \quad I_{\max} = \arg\max_{k=1,\ldots,K} \mathbf{S}_{k,m} \tag{3}$$

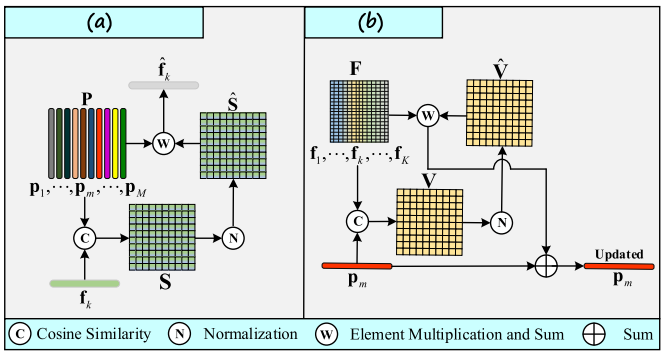


Fig. 3. Process of feature transforming and codebook updating in the BPM module. (a) Transform F. (b) Update P.

$$\hat{\mathbf{f}}_k = \sum_{k=1}^{K} \hat{\mathbf{S}}_{k,m} \mathbf{p}_m. \tag{4}$$

To update $\mathbf{P}$ in Fig. 3(b), we compute the correlation between each prototype items and all feature, and $\mathbf{V}_{k,m}$ is normalized to $\hat{\mathbf{V}}_{k,m}$

$$\mathbf{V}_{k,m} = d(\mathbf{f}_k, \mathbf{p}_m) \tag{5}$$

$$\hat{\mathbf{V}}_{k,m} = \frac{\mathbf{V}_{k,m}}{\mathbf{V}_{I'_{\max}}} \quad I'_{\max} = \arg\max_{m=1,\ldots,M} \mathbf{V}_{k,m} \tag{6}$$

where $\mathbf{V}_{k,m} \in \mathbf{V}$, its size is $K \times M$. The similarity matrix $\hat{\mathbf{V}}$ and input feature $\mathbf{F}$ are used to optimize each prototype item $\mathbf{p}_m$ by

$$\mathbf{p}_m \leftarrow L_2\left(\mathbf{p}_m + \sum_{m=1}^{M} \hat{\mathbf{V}}_{k,m} \cdot \mathbf{f}_k\right) \tag{7}$$

where $L_2(\cdot)$ is the L_2 norm. By the weighted average to the query characteristics, we can assign a higher weight to focus more on the features relevant to the prototype item.

Through insight into the above process, we have found that the prototype codebook construction has three limitations for further improvements.

1) The input $\mathbf{F}$ needs relatively the pure BK samples. Since $\mathbf{P} \in \mathbb{R}^{m\times b}$ is a learnable codebook for constructing various representative BKs, the anomaly pixels mixed with $\mathbf{F}$ can affect the optimization to some extent. To this end, the result of AAMM is employed to provide pure BK samples. The input samples can be transformed into $\mathbf{F} \in \mathbb{R}^{K\times b}(K = N - n)$, where K is the pixel number of BKs and n is the pixel number of anomalies.
2) The initialization of $\mathbf{P}$ plays a crucial role in the BK representation and model convergence. If $\mathbf{P}$ is initialized randomly, the diverse BK features from $\mathbf{F}$ may not be conducive to the BK reconstruction of $\mathbf{X}$ during the early training, necessitating additional iterations for further optimization. We employed the local spatial information center to initialize each prototype item. To achieve this, the mask map and superpixel segmentation map are adopted to obtain the centroid of each local homogeneous region. As for specific implementation process, the principal components analysis (PCAs) is employed to generate the first three principal component features, and the generated features are input into the simple

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linear iterative clustering (SLIC) to obtain a regional division map ($\mathbf{R}$) that removes the potential anomaly by mask map. The number of superpixel blocks of SLIC (η) is set to 100 in this article, because this setting is relatively enough for separating different types of BKs. Drawing on spatial feature invariance, the above spatial information in each local area is applied to $\mathbf{F}$ to generate its corresponding centroid. In this manner, the prototype items serve as the typical BK features in the initialization phase to optimize the model convergence.

3) Inspired by [60], we have designed a comprehensive and effective constraint to transform $\mathbf{F}$ and update $\mathbf{P}$.

A feature compactness loss (L_{FC}) is utilized to minimize the disparity between the input feature vector $\mathbf{f}_k$ and its most-similarity prototype item as defined below

$$L_{\mathrm{FC}} = \sum_{k=1}^{K} \left\| \mathbf{f}_k - \mathbf{p}_m^{\dagger} \right\|_2 \qquad \dagger = \underset{m=\{1,\ldots,K\}}{\arg\max}\ \mathbf{S}_{k,m} \tag{8}$$

where $\|\cdot\|_2$ represents the L_2 norm; $\dagger$ denotes the index of the more relevant prototype item ($\mathbf{p}_m$) for f_k.

To enhance the distinctiveness among $\mathbf{f}_k$, the feature diversity loss (L_{FD}) is introduced as

$$L_{\mathrm{FD}} = \sum_{k=1}^{K} \left(\left\| \mathbf{f}_k - \mathbf{p}_m^{\dagger} \right\|_2 - \left\| \mathbf{f}_k - \mathbf{p}_m^{*} \right\|_2 \right) \qquad * = \underset{m=\{1,\ldots,K\} \cap m \neq \dagger}{\arg\max}\ \mathbf{S}_{k,m} \tag{9}$$

where $*$ denotes the index of the second most relevant prototype item ($\mathbf{p}_m$) for $\mathbf{f}_k$.

To refine each prototype item, a self-attention structure is adopted to mine the relationships among prototype items. This can increase these prototype items discriminative, as defined by

$$\begin{aligned} \mathbf{Q} = \mathbf{p}_m^t \cdot \mathbf{W}_Q,\ \mathbf{K} = \mathbf{p}_m^t \cdot \mathbf{W}_K,\ \mathbf{V} = \mathbf{p}_m^t \cdot \mathbf{W}_V \\ \mathbf{p}_m^{t+1} = \mathrm{Softmax}(\mathbf{Q}\mathbf{K}^T/\sqrt{c})\mathbf{V} + \mathbf{p}_m^t \\ \hat{\mathbf{p}}_m^{t+1} = \mathrm{LayerNorm}\left(fc(\mathbf{p}_m^{t+1}) + \mathbf{p}_m^{t+1}\right) \end{aligned} \tag{10}$$

where $\mathbf{p}_m^t$ denotes the mth prototype item of t epoch; $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$ are the query, key, and value, respectively. It is worth noting that $\mathbf{P}$ is normalized before (10); Softmax$(\cdot)$ is the Softmax function; LayerNorm$(\cdot)$ is the layer normalization; $fc(\cdot)$ represents the function of fully connected layer.

Similar to [61], a prototype separation loss (L_{PS}) is employed to enhance the diversity in the prototype items as defined by

$$L_{\mathrm{PS}} = \sum_{\mathbf{p}_i, \mathbf{p}_j \in \mathbf{P}} \max((\xi - \mathrm{dis}(\mathbf{p}_i,\ \mathbf{p}_j)), 0) \tag{11}$$

where dis$(\cdot)$ denote $\|\cdot\|_2^2$ which is the square of L_2 function; ξ is a preset parameter [61]. L_{PS} is an unsupervised constraint function for the prototype item.

Finally, the combined loss function of BPM is defined as

$$L_{\mathrm{BPM}} = L_{\mathrm{FC}} + \beta L_{\mathrm{FD}} + \gamma F_{\mathrm{PS}} \tag{12}$$

where β and γ denote the weighting coefficients of L_{FD} and F_{PS}. Following [60], the values of β and γ are set to 1, and their settings are rational and effective from the subsequent detection performances.

Spatial–Spectral Interaction Block: The BPM utilizes the interrelation of BK pixels and prototype items by global perspective to optimize the representation of multiclass BKs. Local-window self-attention excels in the HAD but is hindered by a restricted receptive field and inadequate modeling capabilities due to its nonoverlapping window approach and shared weights across channels [62]. Meanwhile, as mentioned in [63], a global-local spatial–spectral awareness had good effectiveness for the HAD. Here, a spatial–spectral interaction block is first utilized for HAD, this block integrates the local-window self-attention with depth-wise convolution in a parallel manner. Its design structure not only enhances the receptive fields by simultaneously modeling intrawindow and cross-window relationships but also achieves bi-directional interactions between the spatial branch and spectral branch. The interactions offset the limits caused by the weight-sharing mechanism and improve the modeling ability in spectral and spatial dimensions by providing complementary clues for local-window self-attention and depth-wise convolution, respectively.

As shown in Fig. 1(a), the SSIB serves as a bridge for enhancing feature representation between the encoder and decoder, and each layer of them has an SSIB. The SSIB structure is displayed in Fig. 1(b) and (c), and the inputs are mask map and $\mathbf{F}_{\mathrm{En}-i}$. $\mathbf{F}_{\mathrm{En}-i}$ denotes the feature of the ith layer of encoder. To construct the local feature interaction, we adopt the $w \times w$ window partition to divide the entire HSI into N_{win} small blocks and reshape their 3-D features into 2-D. Notably, the size of the window is different in SSIB. By the setting, different receptive fields are assigned to explore the relationships between HSI pixels and uncover potential anomalies. In Fig. 1(c), the process of SSIB can be defined by

$$\begin{aligned} \mathbf{F}'_{\mathrm{En}-i} &= \mathrm{Rs}(\mathrm{WP}(\mathbf{F}_{\mathrm{En}-i}, \mathbf{M})) \\ \mathbf{F}_{\mathrm{En-i}}^{\mathrm{DWC}} &= \mathrm{DWC}\left(\mathbf{F}'_{\mathrm{En}-i}\right),\ \mathbf{F}_{\mathrm{En-i}}^{\mathrm{Spe}} = \mathrm{SpeNet}\left(\mathbf{F}_{\mathrm{En-i}}^{\mathrm{DWC}}\right) \\ \mathbf{F}_{\mathrm{En-i}}^{\mathrm{WSA}} &= \mathrm{WSA}\left(Ln\left(\mathbf{F}'_{\mathrm{En}-i}\right),\ \mathbf{F}_{\mathrm{En-i}}^{\mathrm{Spe}}\right) \\ \mathbf{F}_{\mathrm{En-i}}^{\mathrm{Spa}} &= \mathrm{SpaNet}\left(\mathbf{F}_{\mathrm{En-i}}^{\mathrm{WSA}}\right) \\ \mathbf{F}_{\mathrm{En-i}}^{\mathrm{DWC}\prime} &= \mathrm{Conv}\left(\mathbf{F}_{\mathrm{En-i}}^{\mathrm{DWC}}, \mathbf{F}_{\mathrm{En-i}}^{\mathrm{Spa}}\right) \\ \hat{\mathbf{F}}_{\mathrm{En}-i} &= \mathrm{Rs}\left(\mathrm{FNN}\left(\mathrm{Cont}\left(\mathbf{F}_{\mathrm{En-i}}^{\mathrm{DWC}\prime}, \mathbf{F}_{\mathrm{En-i}}^{\mathrm{WSA}}\right)\right)\right) \end{aligned} \tag{13}$$

where $Rs(\cdot)$ and WP$(\cdot)$ represent the operation of reshaping and window-based partition; DWC$(\cdot)$ is the deep-wise convolution, the size of kernel size is set to 3×3 in this article and its setting can get a good balance in accuracy and efficiency [62]; WSA$(\cdot)$, SpeNet$(\cdot)$, and SpaNet$(\cdot)$ are the window-based self-attention function, spectral interaction function, and spatial interaction function, respectively; $Ln(\cdot)$ is the function of layer normalization and Conv$(\cdot)$ is the convolution function with 1×1 kernel size; Cont$(\cdot)$ denotes the concatenation operation according to channel direction; FNN$(\cdot)$ means the feed-forward network based on the fully connected layer, and it can mix the learned relations with channels to generate the

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optimized feature $\hat{\mathbf{F}}_{\mathrm{En}-i}$

$$\begin{aligned}&\mathrm{WSA}\left(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i, \mathbf{F}_{\mathrm{En-i}}^{\mathrm{Spe}}\right)\\&\quad = Rs\left(\mathrm{Softmax}(\mathbf{Q}_i\mathbf{K}_i^{\mathrm{T}} + \mathbf{B})\left(Rs\left(\mathbf{V}_i \otimes \mathbf{F}_{En-i}^{\mathrm{Spe}} v\right)\right)\right)\end{aligned} \tag{14}$$

where $\mathbf{B}$ denotes the relative position bias [62].

In the window-based self-attention, a nonoverlapped window is adopted to accelerate the computational efficiency, though the receptive field is limited for regions which has no cross-window connections. Some HAD methods utilize sliding windows to resolve this issue [49], [63]. In this article, inspired by [64], we adopt a different paradigm by considering connections among windows. Thereby, the depth-wise convolution is an efficient alternative to constructing local relations across connected windows by the parallel design.

The spectral interaction structure, as shown in Fig. 1(c), is based on the SE design [65]. Unlike the attention-based HAD methods, its spectral interaction is not from the current stream, but from another parallel stream. Meanwhile, we solely implement channel interaction within the local window self-attention, rather than applying it to the entire output features. In the spatial interaction part, it is similar to the spectral design. The spatial attention information is from another stream which is the window-based self-attention module. It is combined with the deep-wise convolution with the 3×3 convolutional kernel, this strategy can enhance the utilization of spatial features for the deep-wise convolution branch.

D. Loss Function

Conceding the comprehensiveness in the model optimization, the overall loss function consists of the reconstruction loss (L_{R}) and BPM loss (L_{BPM}), given by

$$\begin{aligned}L &= L_R + \alpha L_{\mathrm{BPM}}\\L_R &= \left\|\mathbf{X}_{i,j} - \hat{\mathbf{X}}_{i,j}\right\|_2\end{aligned} \tag{15}$$

where α is an adjustment coefficient and it is set to 0.1. The training processing of PSSIN is described in Algorithm 1.

Algorithm 1 Training Stage of PSSIN

Input: A HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, AAMM, BPAE (En, BPM, SSIB, De); Parameters: *MinPts*, *Eps*, *LR*, *Epochs*
1: Operate AAMM to generate to the mask map $\mathbf{M}$, Obtain $\mathbf{R}$ by SLIC
2: **While** epoch $\leq$ *Epochs*
3: Initialize $\mathbf{P}$ by $\mathbf{M}$ and $\mathbf{R}$
4: Extract $\mathrm{F}_{\mathrm{En}-1}$, $\mathrm{F}_{\mathrm{En}-2}$, F by En of Eq. (1)
5: Optimize of $\mathbf{P}$ and $\hat{\mathbf{F}}$ by Eq. (2)-(7)
6: Extract $\hat{\mathbf{F}}_{\mathrm{En}-1}$ and $\hat{\mathbf{F}}_{\mathrm{En}-2}$ by Eq. (13) and (14)
7: Generate $\hat{\mathbf{X}}$ by De of Eq. (1)
8: Optimize parameters of BP-AE by Eq. (15), (12), (8), (9) and (11)
9: **End while**
Output: A trained BP-AE

E. Anomaly Detection

The anomaly detection in the PSSIN includes two stages, i.e., the test stage and the evaluation stage. In the test stage, each pixel $\mathbf{X}$ is fed into the trained BP-AE to generate the reconstructed $\mathbf{X}'$ as follows:

$$\mathbf{X'}_i = \mathrm{BP} - \mathrm{AE'}(\mathbf{X}_i, \theta_{\mathrm{BP}-\mathrm{AE'}}) \tag{16}$$

where $\mathrm{BP} - \mathrm{AE'}(\cdot)$ denotes the trained BP-AE function and $\theta_{\mathrm{BP}-\mathrm{AE'}}$ is its parameter.

In the evaluation stage, the anomaly degree is linked to the reconstruction errors as defined by

$$\mathbf{Y}_i = \mathbf{X'}_i - \mathbf{X}_i \tag{17}$$

where $\mathbf{Y}_i$ is the reconstruction error for a pixel.

The Mahalanobis distance is employed to measure the abnormal degree $\mathbf{Y}_i$, defined by

$$\begin{aligned}\hat{\mathbf{Y}}_i &= \mathrm{MD}(\mathbf{Y}_i) = (\mathbf{Y}_i - \mu)^T \Gamma^{-1}(\mathbf{Y}_i - \mu)\\\mu &= \frac{1}{N}\sum_{i=1}^{N}\mathbf{Y}_i,\ \Gamma = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{Y}_i - \mu)(\mathbf{Y}_i - \mu)^T\end{aligned} \tag{18}$$

where $MD(\cdot)$ is the Mahalanobis distance function; μ and $\mathbf{\Gamma}$ are the mean vector and covariance for $\mathbf{Y}$. The anomaly detection processing of PSSIN is shown in Algorithm 2.

Algorithm 2 Anomaly Detection of PSSIN

Input: A HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, A trained BP-AE
1: Generate **X'** by Eq. (16)
2: Calculate $\mathbf{Y}$ by Eq. (17)
3: Obtain $\hat{\mathbf{Y}}$ by Eq. (18)
Output: $\hat{\mathbf{Y}}$

III. Experimental Results and Analysis

A. Experimental Setup

1) HSI Datasets for HAD [9], [10], [66]: For performance evaluation, we have selected six real HSI datasets, namely Gulfport, Cat Island, Pavia, Belcher Bay, Hyperion, and San Diego. They have different spatial sizes, spectral bands, sensors, and spatial resolution, with four types of land features labeled as anomaly targets. The details are shown in Fig. 4 and Table II.

2) Evaluation Index [67]: The 2-D receiver operating characteristic (ROC) curve, 3-D ROC curve, and multiple types of the area under the curve (AUC) are adopted to measure the HAD performance. The 3-D ROC curve is a 3-D curve that describes the changes in detection probability ($\mathrm{P_D}$) and false alarm rate ($\mathrm{P_F}$) at different thresholds (τ), and the 2-D ROC curve is the projection on different surfaces of a 3-D ROC curve. The AUC is the quantitative form of the ROC curve, namely the size of the area enclosed by the ROC curve. The commonly utilized AUCs include $\mathrm{AUC_{TD}}$, $\mathrm{AUC_{BS}}$, $\mathrm{AUC_{ODP}}$, and $\mathrm{AUC_{TDBS}}$, and their formulas are

$$\begin{aligned}\mathrm{AUC_{TD}} &= \mathrm{AUC}_{(D,F)} + \mathrm{AUC}_{(D,\tau)}\\\mathrm{AUC_{BS}} &= \mathrm{AUC}_{(D,F)} - \mathrm{AUC}_{(F,\tau)}\end{aligned}$$

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TABLE II
DETAILED DESCRIPTION OF SIX HYPERSPECTRAL DATASETS

Dataset	Spatial Size	Spectral Bands	Sensor	Spatial Resolution	Anomaly Types
Gulfport	100×100	191	AVIRIS	3.4m	Airplanes
Cat Island	150×150	188	AVIRIS	17.2m	Airplanes
Pavia	150×150	102	ROSIS-03	1.3m	Vehicles
Belcher Bay	150×150	150	GF-5	30m	Ship and Ship tracks
Hyperion	150×150	155	Hyperion	10m	Storage bins and Roofs
San Diego	100×100	189	AVIRIS	3.5m	Airplanes

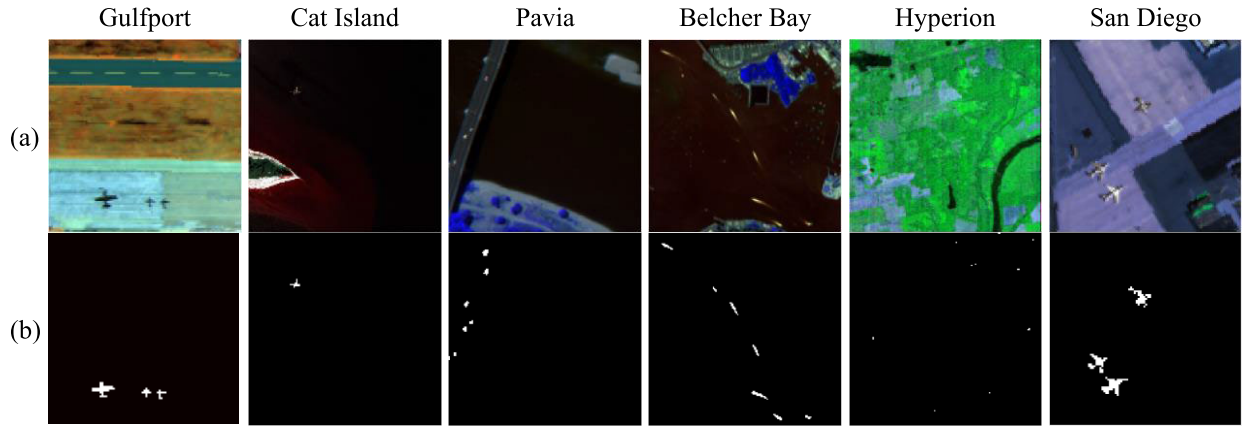


Fig. 4. Six real HSI datasets in this article. (a) False-color image (FCI). (b) Ground-truth (GT).

$$\begin{aligned} \mathrm{AUC}_{\mathrm{TDBS}} &= \mathrm{AUC}_{(D,\tau)} - \mathrm{AUC}_{(F,\tau)} \\ \mathrm{AUC}_{\mathrm{ODP}} &= \mathrm{AUC}_{(D,F)} + \mathrm{AUC}_{(D,\tau)} - \mathrm{AUC}_{(F,\tau)} \end{aligned} \tag{19}$$

where $\mathrm{AUC}_{\mathrm{TD}}$ is used to measure the effect of detecting the anomaly, and $\mathrm{AUC}_{\mathrm{BS}}$ is to evaluate the performance of suppressing the BK; $\mathrm{AUC}_{\mathrm{TDBS}}$ and $\mathrm{AUC}_{\mathrm{ODP}}$ are comprehensive indicators to assess the performance of the anomaly detectability with BK suppressibility.

In addition, the separation map is introduced to measure the separation effect between the anomaly and the BK, and the testing time is utilized to evaluate the detection efficiency.

3) Compared Methods: To validate the superiority of the proposed PSSIN, several representative and advanced algorithms are selected. They are RX [14], CRD [21], NJCR [22], GNLTR [32], AHMID [68], MTVLRR [69], RGAE [39], GAED [44], DFAN [60], MSNet [70], and GTHAD [71] covering different mainstream types of HAD models. Among them, the statistical-based model is RX; CR methods are CRD and NJCR; the LRR types are GNLTR and AHMID; the RGAE, GAED, MSNet, and GTHAD are DL algorithms. The settings of compared methods come from the parameter analysis section of their respective papers, and their detailed settings can be displayed in Table III.

4) Implement Details: The computer utilized in this article is equipped with an Intel i7-6700k CPU at 4 GHz × 8 and an NVIDIA RTX 2080Ti, and the software environment is python-3.7.15 and pytorch-1.13.1. In the settings of SSIB, we provide different receptive fields to dig out the spatial information, and the sizes of SSIB-1 and SSIB-2 are 3 × 3 and 5 × 5. This setting is advantageous in capturing anomalies of different sizes and types. The determination of the remaining parameters is obtained through parameter analysis.

B. Detection Results

In this part, the subjective and objective results are presented to demonstrate the detector performance.

Subjective Results:

As shown in Fig. 5, the detection maps of twelve algorithms are displayed in the six real HSIs. From overall abnormal highlighting performance (highlighting degree and highlighting completeness), the DFAN, GTHAD, and the proposed model are the most outstanding, and the PSSIN has better edge effects than the GTHAD; the NJCR achieves satisfying results; the performances of other algorithms are is not competitive. For example, in the Gulfport and San Diego datasets, the proposed model and DFAN can capture most anomalies and maintain their edges intact; the GTHAD and NJCR can detect most anomaly targets but their edges are not complete enough. Abnormalities of some detectors are either detected in a small part or missed. Especially in the RX, AHMID, and MSNet, the performance of missed detections has become more apparent in the Pavia datasets in which their anomalies are small and difficult to detect. However, the PSSIN, CRD, NJCR GNLTR, RGAE, DFAN, and GTHAD can highlight anomaly objects (vehicles) well.

Significantly, in the DL methods (RGAE, GAED, DFAN MSNet, GTHAD, and PSSIN), PSSIN performance of capturing anomalies is relatively superior to other methods. This result illustrates that the PSSIN design can optimize multiple types of BK land cover and weaken the reconstruction

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TABLE III
PARAMETER SETTINGS OF ELEVEN COMPARED METHODS

Methods	Parameters	Gulfport	Cat Island	Pavia	Belcher Bay	Hyperion	San Diego
RX	\	\	\	\	\	\	\
CRD	(ω_{out},ω_{in},λ)	(13,11,0.01)	(13,11,0.01)	(13,11,0.01)	(17,13,0.1)	(15,11,0.01)	(13,11,0.01)
NJCR	(λ,σ)	(0.001,5)	(0.001,5)	(0.01,5)	(10,5)	(10,5)	(100,5)
GNLTR	(θ,α)	(0.01,0.03)	(0.05,0.001)	(0.005,0.0005)	(0.005,0.0005)	(0.05,0.002)	(0.001,0.05)
AHMID	(α,λ)	(0.0001,0.01)	(0.001,0.01)	(0.001,0.001)	(0.001,0.001)	(0.001,0.1)	(1,0.1)
MTVLRR	(λ,μ)	(0.7,0.2)	(0.7,0.3)	(0.7,0.3)	(0.7,0.9)	(0.7,0.05)	(0.7,0.01)
RGAE	(λ,S,n_hid)	(1×10^{-4},150,100)	(1×10^{-5},50,80)	(1×10^{-5},50,20)	(0.1,50,40)	(0.01,300,60)	(0.01,50,60)
GAED	(c,l_a,β,r_{iter},d)	(3,0.8,1,600,25)	(11,0.4,1,300,25)	(7,0.3,1,400,25)	(11,0.3,1,300,25)	(7,0.3,1,200,25)	(11,0.1,1,200,25)
DFAN	(M,E,Ep,Lr)	(2,0.5,20,1×10^{-4})	(3,0.3,20,2×10^{-4})	(6,0.6,20,1×10^{-4})	(3,0.5,20,5×10^{-3})	(5,0.5,20,1×10^{-4})	(3,0.5,20,2×10^{-4})
MSNet	(λ,Nbs,L)	(1×10^{-4},191,3)	(1×10^{-3},64,3)	(1×10^{-3},64,4)	(1×10^{-3},64,3)	(1×10^{-3},64,3)	(1×10^{-4},64,3)
GTHAD	(*iter*,*lr*)	(300,1×10^{-4})	(100,5×10^{-5})	(100,5×10^{-4})	(100,1×10^{-4})	(100,1×10^{-2})	(100,5×10^{-4})

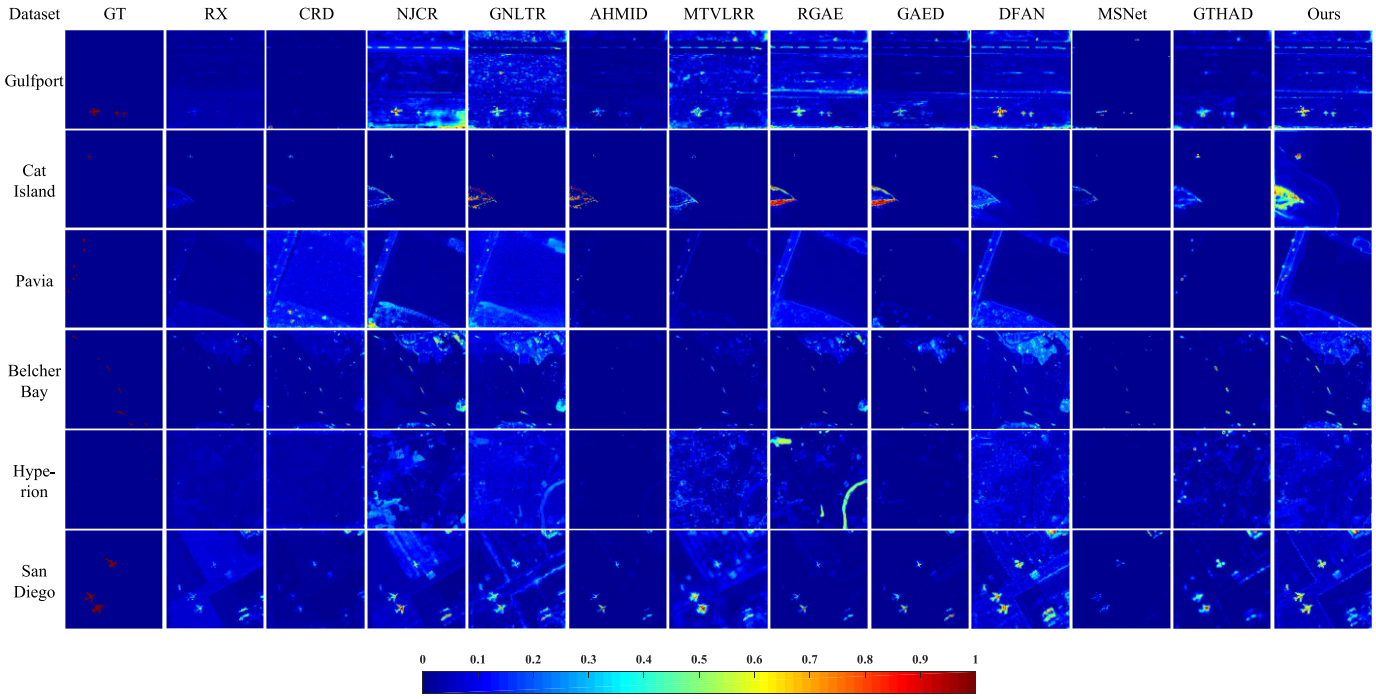


Fig. 5. Detection maps of six real hyperspectral datasets.

of anomaly land cover. As for the performance of alarm false rates, the GTHAD, MSNet, AHMID, and RX have the excellent effect in six HSI datasets, and they can suppress effectively BK. In the Pavia dataset, these three methods and MTVLRR restrain the bridge and land, but their performances of vehicle highlighting are not very ideal; the other detectors generate different levels degrees of alarm false in the BK, the situation of NJCR and GNLTR are serious and the PSSIN performance is acceptable. Compared to SOTA algorithms, the proposed method performs relatively better in terms of comprehensive performance of detection accuracy and false alarm rate.

Objective Results: Objective evaluations include the ROC curves, AUC values, BK-anomaly separability maps, and testing time. The detection performance of these indexes is described as follows.

Fig. 6 compares the 3-D and 2-D ROC curves obtained from our approach and the compared ones. The PSSIN and DFAN demonstrate a higher detection probability P_D across various τ values in the 3-D ROC curves compared to alternative algorithms, and the effect of PSSIN is slightly better than DFAN. To be concrete, the PSSIN detection probability consistently surpasses that of mainstream algorithms across various values of τ, particularly in the Cat Island and San Diego datasets, where the advantage is more significant. However, the proposed model produces some false alarms. This result stems from the removal of some BK in prior knowledge, which to some extent weakens some BK reconstruction capabilities. For instance, in the Belcher Bay dataset, 3-D curves of PSSIN positions itself among the curves of other algorithms, indicating that our method generates some false alarms. The response results from P_D and P_F suggest that achieving high

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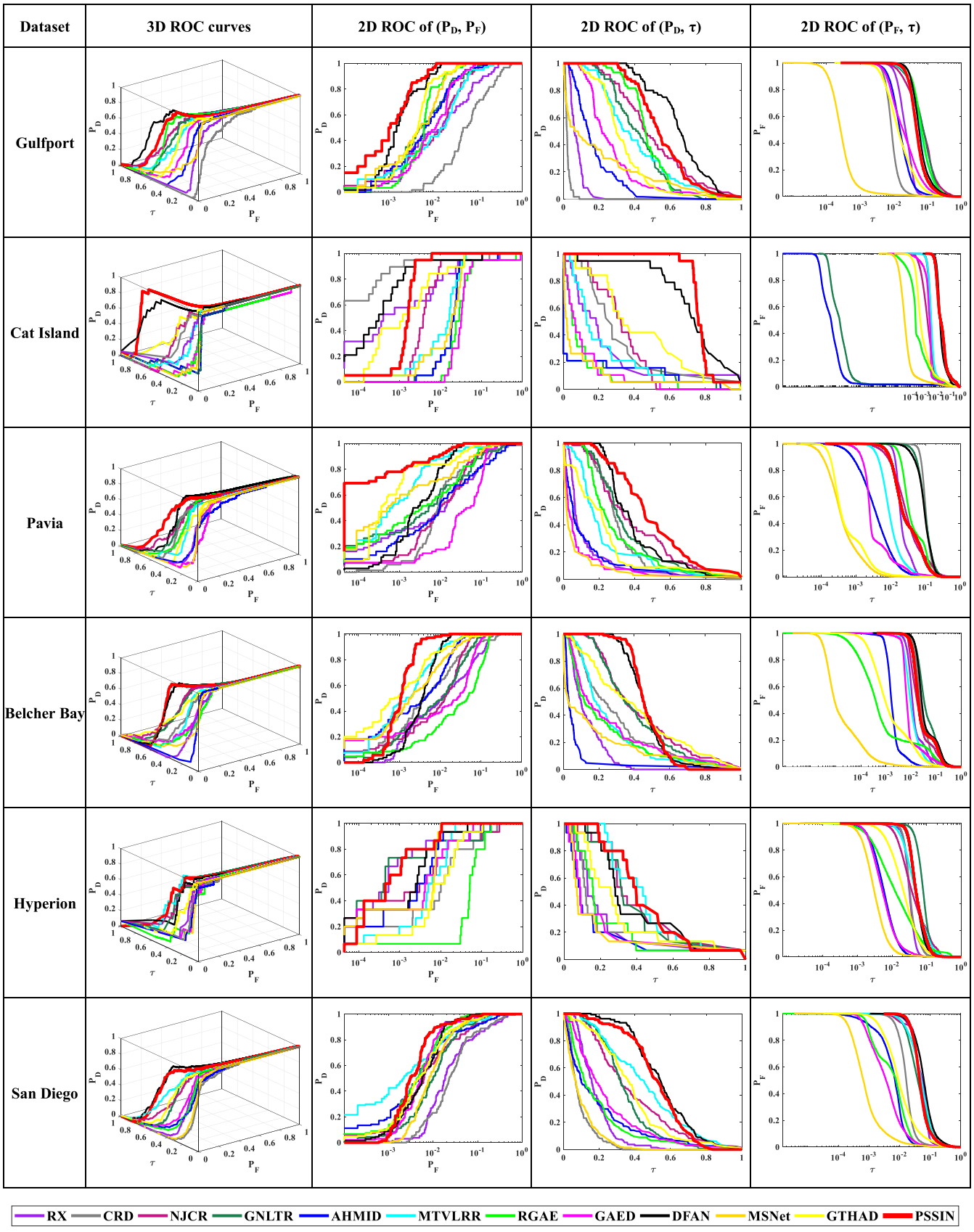


Fig. 6. ROC curves for HAD from six HSI datasets.

detection accuracy while maintaining a low false alarm rate is challenging simultaneously for the unsupervised HAD task.

The 2-D curves for (P_D, P_F), (P_D, τ), and (P_F, τ) represent projections of three distinct axial planes of the 3-D curve.

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TABLE IV

AUC SCORES ON THE SIX REAL HSI DATASETS. THE BOLD ANNOTATIONS DENOTES THE BEST PERFORMANCE AND ANNOTATIONS WITH UNDERLINE MEANS THE SECOND-BEST PERFORMANCE

Dataset	Metrics	RX	CRD	NJCR	GNLTR	AHMID	MTVLRR	RGAE	GAED	DFAN	MSNet	GTHAD	PSSIN
Gulfport	$AUC_{(D,F)}\uparrow$	0.9526	0.8921	0.9760	0.9839	0.9827	0.9720	0.9912	0.9830	0.9969	0.9912	0.9933	**0.9978**
	$AUC_{(D,\tau)}\uparrow$	0.0736	0.0231	0.5034	0.4175	0.1695	0.4064	0.4758	0.2719	**0.6488**	0.1996	0.3305	0.5535
	$AUC_{(F,\tau)}\downarrow$	0.0248	0.0108	0.0984	0.0919	0.0208	0.0822	0.0876	0.0342	0.0756	**0.0020**	0.0257	0.0621
	$AUC_{TD}\uparrow$	1.0262	0.9152	1.4794	1.4014	1.1522	1.3784	1.4670	1.2549	**1.6457**	1.1908	1.3239	1.5513
	$AUC_{BS}\uparrow$	0.9278	0.8813	0.8776	0.8919	0.9619	0.8898	0.9036	0.9488	0.9213	**0.9891**	0.9676	0.9357
	$AUC_{ODP}\uparrow$	1.0015	0.9045	1.3810	1.3094	1.1314	1.2962	1.3794	1.2207	**1.5701**	1.1887	1.2982	1.4892
	$AUC_{TDBS}\uparrow$	0.0489	0.0124	0.4050	0.3256	0.1487	0.3242	0.3882	0.2377	**0.5733**	0.1976	0.3048	0.4914
Cat Island	$AUC_{(D,F)}\uparrow$	0.9807	0.9916	0.9904	0.9688	0.9689	0.9839	0.9352	0.9222	0.9940	0.9806	0.9938	**0.9983**
	$AUC_{(D,\tau)}\uparrow$	0.2496	0.3203	0.3163	0.1207	0.1250	0.2320	0.1059	0.1118	0.7192	0.0989	0.4269	**0.7764**
	$AUC_{(F,\tau)}\downarrow$	0.0065	0.0058	0.0083	0.0073	0.0076	0.0115	0.0143	0.0173	0.0407	**0.0026**	0.0083	0.0598
	$AUC_{TD}\uparrow$	1.2304	1.3119	1.3067	1.0894	1.0939	1.2159	1.0411	1.0340	1.7133	1.0795	1.4207	**1.7746**
	$AUC_{BS}\uparrow$	0.9742	**0.9857**	0.9820	0.9614	0.9613	0.9724	0.9209	0.9049	0.9533	0.9780	0.9855	0.9384
	$AUC_{ODP}\uparrow$	1.2238	1.3060	1.2984	1.0821	1.0863	1.2044	1.0269	1.0167	1.6726	1.0769	1.4124	**1.7148**
	$AUC_{TDBS}\uparrow$	0.2431	0.3145	0.3080	0.1133	0.1174	0.2205	0.0917	0.0944	0.6786	0.0963	0.4186	**0.7166**
Pavia	$AUC_{(D,F)}\uparrow$	0.9538	0.9537	0.9713	0.9616	0.9362	0.9909	0.9771	0.9375	0.9936	0.9808	0.9931	**0.9976**
	$AUC_{(D,\tau)}\uparrow$	0.1343	0.3103	0.3869	0.3531	0.1261	0.2324	0.2966	0.0922	0.3956	0.0750	0.2256	**0.4892**
	$AUC_{(F,\tau)}\downarrow$	0.0233	0.1104	0.0517	0.1062	0.0069	0.0133	0.0557	0.0081	0.1017	**0.0008**	0.0015	0.0437
	$AUC_{TD}\uparrow$	1.0881	1.2640	1.3582	1.3147	1.0623	1.2233	1.2737	1.0296	1.3892	1.0558	1.2187	**1.4868**
	$AUC_{BS}\uparrow$	0.9305	0.8433	0.9196	0.8553	0.9293	0.9775	0.9213	0.9294	0.8920	0.9800	**0.9915**	0.9539
	$AUC_{ODP}\uparrow$	1.0648	1.1536	1.3066	1.2084	1.0554	1.2099	1.2179	1.0215	1.2876	1.0550	1.2171	**1.4432**
	$AUC_{TDBS}\uparrow$	0.1110	0.1999	0.3353	0.2468	0.1193	0.2190	0.2408	0.0840	0.2940	0.0742	0.2241	**0.4455**
Belcher Bay	$AUC_{(D,F)}\uparrow$	0.9622	0.9750	0.9822	0.9781	0.9912	0.9952	0.9407	0.9664	0.9952	0.9904	0.9945	**0.9985**
	$AUC_{(D,\tau)}\uparrow$	0.1294	0.2930	0.4275	0.4007	0.0658	0.2596	0.2276	0.2538	0.4632	0.1640	0.4042	**0.4696**
	$AUC_{(F,\tau)}\downarrow$	0.0191	0.0340	0.0421	0.0635	0.0027	0.0150	0.0188	0.0179	0.0582	**0.0005**	0.0079	0.0452
	$AUC_{TD}\uparrow$	1.0915	1.2680	1.4097	1.3788	1.0571	1.2548	1.1683	1.2203	1.4583	1.1544	1.3987	**1.4681**
	$AUC_{BS}\uparrow$	0.9430	0.9410	0.9402	0.9146	0.9885	0.9803	0.9219	0.9485	0.9369	**0.9899**	0.9866	0.9532
	$AUC_{ODP}\uparrow$	1.0724	1.2340	1.3676	1.3154	1.0544	1.2398	1.1495	1.2024	1.4001	1.1539	1.3908	**1.4229**
	$AUC_{TDBS}\uparrow$	0.1102	0.2590	0.3854	0.3373	0.0631	0.2446	0.2088	0.2359	0.4049	0.1636	0.3963	**0.4244**
Hyperion	$AUC_{(D,F)}\uparrow$	0.9896	0.9675	0.9718	0.9845	0.9858	0.9920	0.9367	0.9887	0.9945	0.9950	0.9821	**0.9978**
	$AUC_{(D,\tau)}\uparrow$	0.2327	0.2466	0.4133	0.3940	0.1761	**0.4439**	0.2352	0.2040	0.3907	0.1857	0.3165	0.4438
	$AUC_{(F,\tau)}\downarrow$	0.0331	0.0384	0.0623	0.0858	0.0081	0.0649	0.0418	0.0086	0.0575	**0.0048**	0.0336	0.0607
	$AUC_{TD}\uparrow$	1.2223	1.2140	1.3851	1.3784	1.1619	1.4359	1.1719	1.1927	1.3852	1.1808	1.2986	**1.4416**
	$AUC_{BS}\uparrow$	0.9566	0.9291	0.9095	0.8986	0.9777	0.9566	0.8949	0.9801	0.9370	**0.9902**	0.9486	0.9371
	$AUC_{ODP}\uparrow$	1.1893	1.1756	1.3228	1.2926	1.1538	1.3710	1.1301	1.1841	1.3277	1.1759	1.2650	**1.3809**
	$AUC_{TDBS}\uparrow$	0.1996	0.2082	0.3509	0.3081	0.1681	0.3790	0.1934	0.1954	0.3332	0.1809	0.2829	**0.3831**
San Diego	$AUC_{(D,F)}\uparrow$	0.9403	0.9412	0.9754	0.9764	0.9732	0.9909	0.9909	0.9877	0.9879	0.9820	0.9855	**0.9916**
	$AUC_{(D,\tau)}\uparrow$	0.1778	0.0911	0.3818	0.3425	0.1930	0.4537	0.1888	0.2386	0.5108	0.0939	0.3734	**0.5140**
	$AUC_{(F,\tau)}\downarrow$	0.0589	0.0214	0.0655	0.0592	0.0094	0.0646	0.0107	0.0094	0.0705	**0.0038**	0.0241	0.0613
	$AUC_{TD}\uparrow$	1.1181	1.0323	1.3572	1.3189	1.1662	1.4365	1.1797	1.2262	1.4987	1.0759	1.3589	**1.5056**
	$AUC_{BS}\uparrow$	0.8814	0.9199	0.9099	0.9172	0.9639	0.9181	**0.9801**	0.9783	0.9174	0.9782	0.9613	0.9303
	$AUC_{ODP}\uparrow$	1.0592	1.0110	1.2917	1.2597	1.1569	1.3718	1.1690	1.2169	1.4281	1.0721	1.3347	**1.4443**
	$AUC_{TDBS}\uparrow$	0.1189	0.0698	0.3163	0.2832	0.1837	0.3891	0.1781	0.2292	0.4402	0.0902	0.3493	**0.4527**

As for ROC curves of (P_D, P_F), the PSSIN occupies the upper left corner of the graph, with our detection probability values at most points exceeding those of other algorithms across all hyperspectral datasets. The DFAN achieves suboptimal results in Gulfport, Cat Island, Pavia, and Belcher Bay datasets, and MSNet secures the second-best outcome in the Hyperion dataset. Conversely, RX, CRD, NJCR, GNTLR, and AHMID show less impressive performance in the (P_D, P_F) ROC curves. Similarly, in the (P_D, τ) ROC curves, the PSSIN and DFAN exhibit higher detection probabilities across most HAD datasets at varying threshold values. Furthermore, when the PSSIN is compared to DL algorithms, our results stand out in the (P_D, τ) ROC curves. Additionally, AHMID, GTHAD, and MSNet clearly yield fewer false alarms than other methods in the (P_F, τ) ROC curves.

The AUC score quantitatively represents the ROC curve, with the AUC results for six HAD datasets detailed in Table IV. We categorize these six HSI datasets into two types: small anomaly target type and large anomaly target type.

For analysis, we utilize the Hyperion dataset as an example of the small anomaly target type. Our AUC scores for $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and AUC_{TD} are 0.9985, 0.4696, and 1.4681, respectively; and the proposed model demonstrates competitive advantages in anomaly detectability. The MSNet

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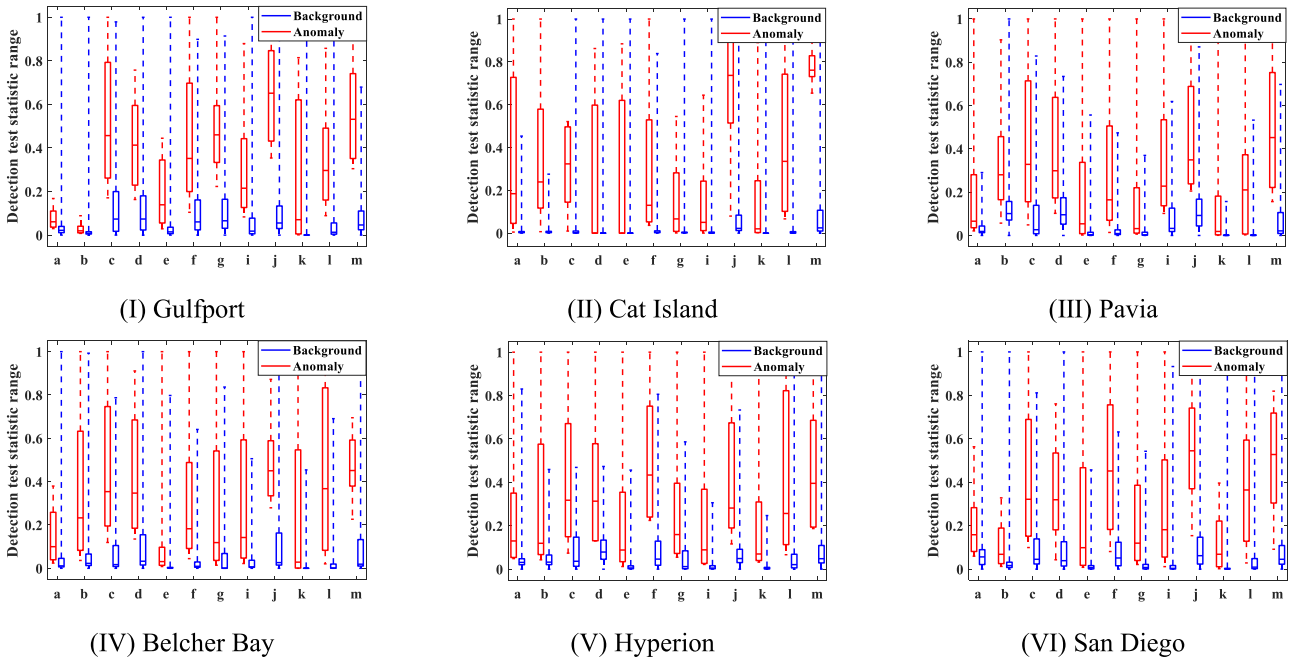


Fig. 7. Separability maps. In each dataset a: RX, b: CRD, c: NJCR, d: GNLTR, e: AHMID, f: MTVLRR, g: RGAE, i: GAED, j: DFAN, k: MSNet, l: GTHAD, and m: PSSIN. (I) Gulfport. (II) Cat island. (III) Pavia. (IV) Belcher Bay. (V) Hyperion. (VI) San Diego.

TABLE V
COMPARING THE TESTING TIME ON THE TEN ALGORITHMS IN SECONDS (S), THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BOLD OR UNDERLINED

	RX	CRD	NJCR	GNLTR	AHMID	MTVLRR	RGAE	GAED	DFAN	MSNet	GTHAD	PSSIN
Gulfport	0.316	7.224	6.3676	3.582	23.303	106.435	0.021	0.044	0.117	**0.002**	0.033	0.135
Cat Island	0.309	41.218	14.894	15.488	46.371	140.421	0.045	0.093	0.257	**0.002**	0.061	0.324
Pavia	0.184	14.861	17.049	15.364	34.99	43.392	0.030	0.081	0.155	**0.002**	0.055	0.219
Belcher Bay	0.323	39.376	14.438	14.902	40.939	129.233	0.041	0.094	0.203	**0.002**	0.077	0.264
Hyperion	0.228	26.352	15.586	13.336	53.569	113.945	0.038	0.092	0.227	**0.002**	0.063	0.268
San Diego	0.178	22.991	15.342	5.317	19.473	30.858	0.022	0.052	0.119	**0.002**	0.032	0.141
Average	0.256	25.337	13.946	11.332	36.441	94.047	0.033	0.076	0.180	**0.002**	0.054	0.225

achieves suboptimal performance with an $AUC_{(D,F)}$ score of 0.9950, while the MTVLRR model ranks first in $AUC_{(D,\tau)}$ and its AUC_{TD} is the second-best effect. In terms of BK suppressibility, MSNet excels with the highest $AUC_{(F,\tau)}$ value and AUC_{BS} score, and their values are 0.0048 and 0.9902. Our algorithm shows performance in the joint evaluation of anomaly detectability and BK suppressibility, with AUC_{ODP} and AUC_{TDBS} scores of 1.3809 and 0.3831, respectively.

With respect to the type of big target anomalies, we can utilize the Gulfport dataset as a typical case to illustrate the detection effect of ten algorithms. Our detector demonstrates ideal performance across the three metrics of anomaly detectability: $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and AUC_{TD}. Similar to the Hyperion dataset, the DFAN detector achieves the best scores for $AUC_{(D,\tau)}$ and AUC_{TD}, while PSSIN leads in $AUC_{(D,F)}$. MSNet records the lowest $AUC_{(F,\tau)}$ values and AUC_{BS} results across all models. The DFAN exhibits the best joint performance in detecting anomalies and suppressing BKs; the PSSIN obtains the second-best performance and the scores of AUC_{ODP} and AUC_{TDBS} are 1.4892 and 0.4914.

Overall, the AUC results from the six real HSI datasets indicate that PPSIN achieves the best performance in anomaly detectability. However, the BK suppressibility effect remains limited for the proposed method. When considering both the anomaly detectability and BK suppressibility, PSSIN algorithm performs relatively well compared to all SOTA algorithms. Notably, when we compare the AUC results of other DL models, we observe that the proposed model outperforms them across six HSI datasets. This improvement arises from the BPM, which enhances the representation of various BK land covers. Meanwhile, the SSIB employs different receptive fields to extract local spatial information. However, the aforementioned algorithms primarily treat the BK as one type of feature, which somewhat restricts their capacity to accurately reconstruct the BK.

The separability map assesses the isolation of BKs and anomalies, and the results of twelve methods are displayed in Fig. 7. The blue box indicates the numeric range of all BK pixels. Similarly, the red box highlights the range of pixel values associated with anomalies. Both boxes contain median lines, which represent the average values of the respective pixel

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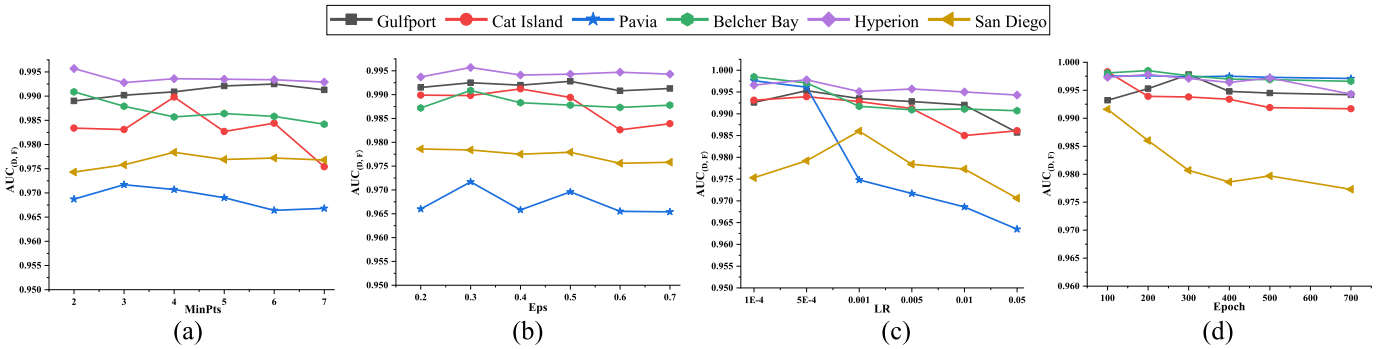


Fig. 8. Results of parameter analysis in the proposed model. (a) MinPts. (b) Eps. (c) LR. (d) Epochs.

TABLE VI
TRAINING TIME ON THE DL ALGORITHMS

Dataset	RGAE	GAED	DFAN	MSNet	GTHAD	PSSIN
Gulfport	41.554	167.956	19.834	**11.859**	263.753	27.401
Cat Island	212.262	106.804	99.317	**11.039**	378.272	24.360
Pavia	207.213	76.629	130.259	**11.797**	212.343	36.447
Belcher Bay	172.928	87.033	138.635	**10.419**	303.406	37.130
Hyperion	167.589	53.869	101.708	**12.088**	314.011	35.137
San Diego	79.103	35.726	22.870	**7.338**	87.810	10.938
Average	146.775	88.003	85.437	**10.757**	259.932	28.569

TABLE VII
COMPUTATIONAL RESOURCE CONSUMPTION OF THE PSSIN

Dataset	Training Phase		Testing Phase	
	CPU Memory	GPU Memory	CPU Memory	GPU Memory
Gulfport	2877.73M	310.74M	52.54M	307.33M
Cat Island	3377.61M	664.83M	2.65M	657.55M
Pavia	2874.57M	391.48M	62.27M	384.48M
Belcher Bay	2895.87M	551.23M	40.5M	544.18M
Hyperion	2896.59M	559.06M	93.95M	550.29M
San Diego	2874.11M	309.71M	52.66M	305.24M
Average	2966.08M	464.51M	50.76M	458.18M

counts. As shown in the blue box results of twelve detectors, AHMID, GAED, and MSNet maintain values within a narrow range across most HSI datasets, suggesting their effectiveness in BK suppression; NJCR, GNLTR, RGAE, PSSIN have a long BK box range, which indicates that these methods have limited BK suppression ability; the BK suppression performance of RX, CRD and GTHAD are at a moderate level in six HSI datasets. Taking Hyperion dataset as an example in Fig. 7(V), the range of blue boxes in the NJCR, GNLTR, MTVLRR, RGAE, DFAN, and PSSIN is relatively long; the detectors of AHMID, GAED, and MSNet have short blue boxes. However, these detectors exhibit limited performance in distinguishing BKs from anomalies. As shown in Fig. 7(III), the overlap between the blue and red boxes in the AHMID, GAED, and MSNet indicates unsatisfactory isolation result of BKs and anomalies. Conversely, the separability maps for NJCR, MTVLRR, DFAN, GTHAD, and PSSIN effectively increase the distance between the red and blue boxes, with our method and MTVLRR demonstrating the most significant effect. For instance, the PSSIN detector maintains a narrow range for both BK and anomaly intervals in the Cat Island dataset, resulting in a pronounced separation effect. The superior performance of the separability map underscores the advancement of PSSIN design.

We evaluate the model efficiency by comparing the testing time and training time, with results presented in Tables V and VI, respectively. Traditional methodologies, owing to their integrated optimization and detection phases, are solely assessed based on testing time. Conversely, DL paradigms, characterized by training and inference stages, facilitate the evaluation of detection efficacy through the metrics of training and testing time. The algorithms are categorized into two groups: traditional algorithms and DL algorithms. Among the traditional algorithms (RX, CRD, NJCR, GNLTR, and AHMID), RX demonstrates a straightforward and effective design, achieving the fastest detection speed with an average inference time of 0.256 s across six HSI datasets. GNLTR follows as the second most efficient, with an average detection time of 11.332 s, attributed to the more streamlined optimization strategies. In the testing time of DL detectors, PSSIN and DFAN require additional time to identify anomalies in each HSI dataset, the average testing time is 0.225 s in the PSSIN and the average testing time is 0.180 s. This is due to the utilization of Mahalanobis distance for assessing the anomaly degree, which increases computational demands. The MSNet construction is relatively simple, and it achieves the highest detection efficiency. Meanwhile, the training time is employed to evaluate the complexity of the DL model design. In the results of Table VI, the MSNet also obtain optimal performance and the proposed network gets suboptimal, and their average training time are 10.757 and 28.569. However, other DL methods need more time to train their models because their designs are relatively complex. To be concrete, RGAE utilizes the graph theory to optimize model; GAED employs the curvature filtering based on local windows; DFAN adopts complicated optimization design; GTHAD leverages the content similarity and gating unit to guide HSI reconstruction.

To this end, these ways require more calculations to implement. Given the superior detection accuracy of the proposed PSSIN, its inference time remains acceptable compared to DL methods. However, PSSIN has a competitive advantage in the training time.

Furthermore, the resource utilization of PSSIN is presented in Table VII. Analysis reveals that the CPU memory footprint

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TABLE VIII
ABLATION STUDY EXPERIMENT. THE $AUC_{(D,F)}$ IS ADOPTED TO MEASURE DETECTION PERFORMANCES

Module		Real HSI datasets						Average
BPM	SSIB	Gulfport	Cat Island	Pavia	Belcher Bay	Hyperion	San Diego	
		0.9566	0.9785	0.9527	0.9631	0.9847	0.9469	0.9638
√		0.9909	0.9917	0.9894	0.9806	0.9942	0.9730	0.9867
	√	0.9915	0.9931	0.9908	0.9906	0.9908	0.9768	0.9889
√	√	**0.9978**	**0.9983**	**0.9976**	**0.9985**	**0.9978**	**0.9916**	**0.9969**

across the six HSI datasets approximates 3000 M during the training phase, whereas the CPU memory consumption for each HSI dataset remains below 100 M during the testing phase, indicating substantially elevated computational demands during training. This is because the PSSIN's training stage incorporates DBSCAN and SLIC, thereby contributing to considerable computational complexity. Conversely, the testing phase of PSSIN employs a simplified Mahalanobis distance. PSSIN requires a certain amount of GPU memory, primarily attributed to the intricate model architecture of BP-AE. In summary, the implementation of PSSIN necessitated the integration of reasonable prior knowledge and the construction of a relatively complex neural network to facilitate effective PSSIN detection.

C. Ablation Study

The proposed model is founded on the AE, incorporating BPM and SSIB to enhance the detection performance in this article. Consequently, the detection accuracy is not ideal in the absence of the BPM and SSIB, with an average AUC score of 0.9638 across six HSI datasets. Upon integrating the BPM into the AE, we observe a substantial improvement in performance across all datasets, particularly in the Gulfport dataset, where detection accuracy increases by 3.43%. In comparison to the baseline, the model incorporating the BPM achieves an average AUC value that is enhanced by 2.29%. These findings suggest that the BPM design effectively enhances the representation of various BKs. The introduction of the SSIB in the baseline contributes to an improved detection rate, indicating that its design is advantageous for leveraging spatial and spectral information to effectively identify anomalies. As illustrated in Table VIII, the model that integrates both the BPM and SSIB achieves a further enhancement in detection accuracy, culminating in an average AUC score of 0.9969. Thus, the above results substantiate efficacy of the PSSIN overall design.

D. Parameter Analyses

The PSSIN incorporates four parameters, MinPts, Eps, LR, and Epochs. The MinPts and Eps are the parameters of DSCAN and they significantly influence the effectiveness of the abnormal mask. The MinPts represents the minimum number of points, with values set to [2], [3], [4], [5], [6], and [7]. As illustrated in Fig. 8(a), the overall $AUC_{(D,F)}$ curves exhibit an initial upward trend followed by a downward trend. The remaining datasets achieve the best $AUC_{(D,F)}$ scores at the MinPts value of 2, with their subsequent curves either fluctuating within a narrow range or gradually decreasing. The Eps denotes the minimum distance between adjacent points and its analysis results are depicted in Fig. 8(b), and its range is 0.2–05 and the interval between two successive values is 0.1. Six curves attain the highest detection accuracy in the mid-range settings of 0.3, 0.4, and 0.5. This finding indicates that excessively large or small Eps settings adversely affect the clustering of abnormal pixels. Fig. 8(c) shows the impact of the learning rate and the settings are [0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05]. Except for the San Diego dataset, all other datasets are more conducive to model optimization and improve the detection effect when the learning rate is relatively small. The AUC curve for the San Diego dataset exhibits a unimodal shape, with the optimal value identified at 0.001. The Epochs settings in Fig. 8(d) are [100, 200, 300, 400, 500,600, 700]. In the San Diego dataset, $AUC_{(D,F)}$ values experience a rapid decline in the first half, followed by fluctuations within a certain range in the latter half. The five other datasets show minor fluctuations, with optimal results concentrated at either 100 or 200 epochs. The optimal parameter settings are presented in Table IX.

TABLE IX
DETAILS OF FOUR HYPERPARAMETERS IN THE PSSIN

Datasets	Gulfport	Cat Island	Pavia	Belcher Bay	Hyperion	San Diego
MinPts	6	4	3	2	6	4
Eps	0.5	0.4	0.3	0.3	0.3	0.3
LR	5×10^{-4}	5×10^{-4}	1×10^{-4}	1×10^{-4}	5×10^{-4}	1×10^{-3}
Epochs	300	100	200	200	200	100

E. Discussion

To validate the efficacy and generalizability of the proposed PSSIN, we conduct a comparative analysis against SOTA methods utilizing HSI datasets characterized by distinct anomaly targets. As depicted in Fig. 9, the scene, acquired from Salisbury Valley, California, USA, presents small, challenging-to-detect anomalies. Both MSNet and PSSIN demonstrate superior anomaly detection capabilities, achieving $AUC_{(D,F)}$ scores at 0.9954 and 0.9984, respectively, while the other compared algorithms exhibit suboptimal performance. In Fig. 10, the HYDICE dataset scene, captured in an urban area of California, USA, features relatively complex BKs. Despite this, most detectors achieve commendable results, with

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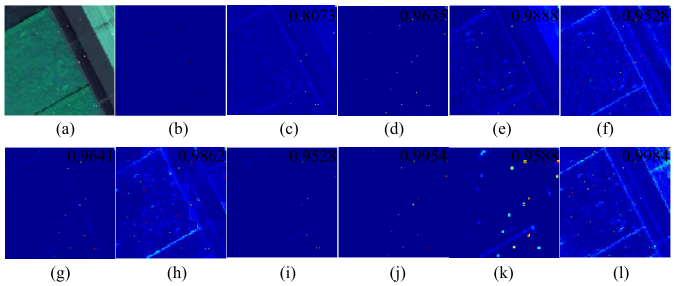


Fig. 9. Detection results of twelve detectors in the Salinas dataset, and the anomaly type is the simulated targets. The red mark means the $AUC_{(D,F)}$. (a) FCI. (b) GT. (c) RX. (d) CRD. (e) NJCR. (f) GNLTR. (g) AHMID. (h) RGAE. (i) GAED. (j) MSNet. (k) GTHAD. (l) PSSIN.

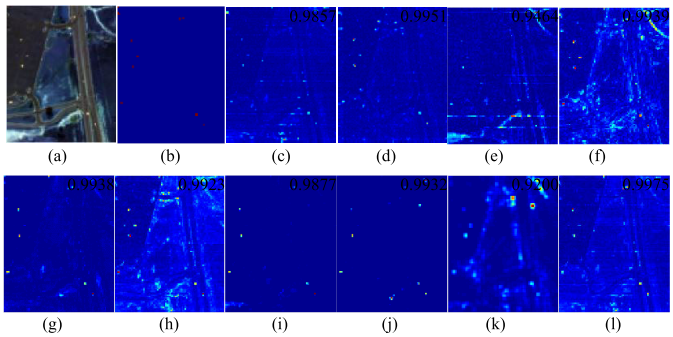


Fig. 10. Detection results of twelve detectors in the HYDICE dataset, and anomaly objects are roofs and cars. (a) FCI. (b) GT. (c) RX. (d) CRD. (e) NJCR. (f) GNLTR. (g) AHMID. (h) RGAE. (i) GAED. (j) MSNet. (k) GTHAD. (l) PSSIN.

AUC values for CRD, GNTLR, AHMID, RGAE, MSNet, and PSSIN recorded at 0.9951, 0.9939, 0.9938, 0.9923, 0.9932, and 0.9975, respectively. Overall, the PSSIN demonstrates a competitive advantage among twelve evaluated methods.

IV. Conclusion

In this article, we propose a novel HAD model, PSSIN, which combines a prototype learning strategy with a spatial–spectral paradigm to construct the AE framework. The BPM based on a multivariate loss function can quantify typical BK features by a global perspective to effectively improve the problem of multiple BK representation. SSIB develops a spatial–spectral bi-directional interaction paradigm for HAD, as a plug-and-play module to enhance the spatial–spectral utilization. Ablation studies and comparison experiments have validated the rationality and superiority of the proposed PSSIN. However, from both subjective and objective results, the proposed method can be improved in terms of the false alarm rate. Therefore, denoising and quality improvement models [72], [73] as well as semi-supervised learning [74] can be explored. Based on the representation of diverse BKs and spatial–spectral interactions, how to build a network and construct more reasonable priors to better cope with complex and diverse scenarios can also be focused in our future work.

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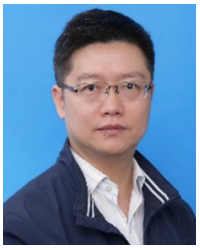
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